

Building Our Digital World: Computer Systems & Architecture

COMP1860

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Activity Sheet 1.1

1 Mechanical Devices

1.1 Antikythera Mechanism

1. When does the Antikythera Mechanism date to?

Solution: It has been dated to somewhere between 205 BC and 60 BC (latest date the ship it was found on sank).

2. What was it used to calculate?

Solution: It could calculate astronomical positions: the sun and moon, moon phase and eclipses.

3. (*) Recent research (27 June 2024) has corroborated a hypothesis about the rings in the device. What is this hypothesis and which UK university corroborated it?

Solution: That the rings are based on a lunar calendar of 354 days rather than the Egyptian solar calendar as previously thought. This was conjectured by Budiselic et al. in 2020 (British Horological Institute) and then corroborated by a team at Glasgow University.

1.2 Al-Jazari's Castle Clock

1. When does it date to?

Solution: 1206 as described in Al-Jazari's *Kitab fi ma'rifat al-hiyal al-handasiya* (*The Book of Knowledge of Ingenious Mechanical Devices*).

2. What powered the clock?

Solution: It was a water powered device.

3. (*) It has been described as the earliest programmable computer? How could it be programmed?

Solution: The length of the day and night could be set. Hence differing inputs would produce different outputs. Full details on its settings are detailed at <https://muslimheritage.com/al-jazaris-castle-water-clock/>

1.3 Charles Babbage's Analytical Engine

1. When does it date to?

Solution: 1837.

2. It has been described as the first general purpose computer. What similarities to modern computers did it have?

Solution: It has a 'Store' similar to registers and memory, the 'Mill' which handled arithmetic processing similar to a CPU, serial operation using a 'fetch-execute' cycle, and input (punch cards) and output (graph plotting, punch cards, print outs).

3. The Analytical Engine could be programmed from instructions loaded into the machine. What was the method used to do this? What earlier machine was this method inspired by?

Solution: It used punch cards. It was inspired by the Jacquard Loom.

4. (*) Who is credited as the first computer programmer? What did this person show the Analytical Engine could calculate by detailing the steps it would undertake to perform this calculation?

Solution: Ada Lovelace. Note G, an appendix to Lovelace's translation of Charles Babbage's lecture on the analytical engine at the University of Turin contained a description of the steps that the Analytical Engine would use to compute Bernoulli numbers. It would be called an "execution trace" today but demonstrated how the Analytical Engine could be programmed to compute a complex function.

2 Electromechanical Devices

2.1 Konrad Zuse's Z3

1. When does it date to?

Solution: It was designed in 1938 and completed in 1941.

2. What was the principal component of its design? How many were used?

Solution: Electrical relays (a switching device). 2600 relays formed part of its design.

3. The Z3 was the first binary computer. What word size did it use?

Solution: 22-bit.

4. (*) Which US mathematician (a seminal figure in computer science) also showed that electrical relays could be used to create Boolean circuits. When was this published?

Solution: Claude Shannon published this in his 1937 Master's thesis "A Symbolic Analysis of Relay and Switching Circuits". Zuse appeared to have been unaware of this and developed his ideas independently.

3 Electronic Devices

3.1 Colossus

Colossus is regarded as the first programmable electronic digital computer.

1. When does it date to?

Solution: 1943-45.

2. What was it used for?

Solution: Cryptanalysis of the Lorenz cipher.

3. What was the component used to perform Boolean logic and arithmetic operations?

Solution: Thermionic valve.

4. Who were its designers?

Solution: Tommy Flowers and Max Newman.

5. (*) What did the US National Security Agency use Colossus for after WWII?

Solution: They used it check one-time pads for non-randomness.

3.2 Manchester Baby

The Manchester Baby was the first stored-program electronic computer.

1. When does it date to?

Solution: 1948.

2. What arithmetic operations could it perform in hardware?

Solution: Subtraction and negation.

3. What did the binary instruction 000 do?

Solution: Jump to the instruction at the specified address.

4. (*) What was the name of the commercial computer that was a direct successor of the Baby?

Solution: Ferranti Mark 1

3.3 Transistor Computers

1. At which UK University was the first experimental transistor computer devised?

Solution: University of Manchester.

2. What was the name of MIT's 1956 experimental transistor computer? What company did the designers go on to form?

Solution: TX-0. DEC (Digital Equipment Corporation).

3. What was the name of the first commercial fully transistorised calculator?

Solution: IBM 608.

3.4 Integrated Circuits

The creation of metal-oxide semiconductor field-effect transistors (MOSFET) in 1960 at Bell Laboratories gave rise to very large-scale integration (VLSI). General Microelectronic created the first commercial MOSFET integrated circuit (IC).

1. When was it released?

Solution: 1964.

2. How many transistors were on the chip?

Solution: 120.

3.5 Mainframe Computers

Mainframe computers were the earliest form of commercial computer. These machines were large and the components typically filled a room. The IBM System/360 was a popular mainframe computer.

1. When was it first released?

Solution: 1964.

2. What was its word size?

Solution: 32-bit.

3. How many registers did its processor have?

Solution: 16.

3.6 Minicomputers

Minicomputers were much smaller than their mainframe predecessors. The entirety of the hardware could now be contained in a cabinet.

1. Which well-known operating system was first designed to run on the PDP-7 minicomputer?

Solution: UNIX.

2. (*) The VAX minicomputer used its own native operating system. What was it called?

Solution: VMS

3.7 Personal Computers

In the 1970s, computing devices could be built reliably and cheaply enough that they were now affordable for individuals. The first commercially successful personal computer was released in 1974.

1. Which company produced it and what was it called?

Solution: Micro Instrumentation and Telemetry Systems (MITS) Altair.

2. What CPU did it use?

Solution: Intel 8080.

3. (*) Which well-known software company produced the programming language for it?

Solution: Microsoft created Altair BASIC.

3.8 Modern CPUs

3.8.1 x86

x86 describes the architecture of many CPUs today.

1. What was the original processor of this family?

Solution: Intel 8086.

2. When was it released?

Solution: 1978.

3. (*) Two instruction set extensions are provided by x86 chip manufacturers to support hardware-assisted virtualization. What are they called?

Solution: VT-x (Intel) and AMD-V (AMD)

3.8.2 ARM

ARM is another popular CPU architecture.

1. When was ARM introduced?

Solution: 1985.

2. Is its design CISC or RISC?

Solution: RISC.

3. One of its principal designers is a Professor at a UK university. What is his name and which University is he associated with?

Solution: Professor Steve Furber. University of Manchester.

4. (*) Give the name of a recent ARM based CPU or SoC (system on chip)?

Solution: Apple A and M series, Qualcomm Snapdragon.

3.8.3 CPUs

1. (*) For the following recent processors:
- Categorise each as a CISC or RISC based architecture.
 - Give the number of transistors on the chip (reported or estimate).
- (a) Intel Core i9-13900K
Solution: CISC. Approximately 26 billion transistors.
- (b) Apple M2 Max
Solution: RISC. 67 billion transistors.
- (c) AMD Epyc Bergamo
Solution: CISC. 82 billion transistors.

4 Computer Architecture

1. Explain what is meant by *architecture* and *organisation* with respect to computer hardware?
Solution: *Architecture* is the logical design of a computer and *organisation* is the physical implementation of that design. An architecture might define what instructions the computer uses, such as a multiplication instruction, whereas an organisation might define whether a specific multiplication unit is required to implement this.
2. List the common functions of computers?
Solution: The common functions are data processing, data storage, data movement and control.
3. Explain why a *hierarchical structure* is used to design computers?
Solution: A hierarchical structure is used to make implementation easier. In a hierarchical structure, each component has fewer dependencies, they only depend on components lower in the hierarchy. This means one can start with simple easy to implement components with no dependencies, then use those simple components to build more complex components, then use those more complex components to build new components and so on.
4. Describe the four components of the *von Neumann* architecture?
Solution: The components are main memory, which stores programs and data, the arithmetic and logic unit, which performs arithmetical and logical operations on data, a control unit, which interprets instructions and executes them, input and output equipment.
5. Explain how the *von Neumann* and *Harvard* architectures differ?
Solution: The *von Neumann* architecture uses a single memory bus for programs and data. The *Harvard* architecture has separate memory buses for programs and data.
6. Describe and explain the fundamental building blocks components of a computer?
Solution: The fundamental components are the memory cell, which stores a bit, and the logic gate which performs one logical function.

7. Describe and explain the fundamental electronic building block of a computer?
Solution: The fundamental electronic component is the transistor, a three input electronic device. It has an input, output and gate. If a voltage is applied to the gate, it allows current to flow from the input to the output. This means the transistor acts as an electronic switch
8. Define Moore's Law?
Solution: Moore's Law states the observation that the number of transistors on a chip has roughly doubled every 18 months.
9. Describe what an instruction set architecture defines?
Solution: An instruction set defines the instructions that a computer will use, what registers are required, how data is represented in memory, hardware support for memory, and the input/output model.
10. Explain the difference between Complex Instruction Set Computer (CISC) and Reduced Instruction Set Computer (RISC)?
Solution: A CISC machine has instructions for different functions. As more CPU functions are introduced, new instructions are added to support these functions. CISC machines have complexity in their set of instructions. A RISC machine has a very simple instruction set defining basic operations like fetching data from and loading data into registers, simple arithmetic and logical operations. Any more complex functionality is defined using these simple instructions. RISC machines have complexity in compiler design which must translate complex programs into the simple RISC instructions.